



ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions –both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

Carrying the tablet one-handed while managing patients/equipment raises drop and collision risk - worse late in a night shift.		Secure strap/holster so it's never the only point of contact; lightweight, grippable case for one-handed use.	
PROBLEM	USER	ROBOT	SOLUTION
DEVICE	ROBOT	DEVICE	
Drops are routine, not an edge case, across corridors, bedsides, and desks.		Drop-rated ruggedised case + screen protector; manual fallback triggers cleanly if the device fails.	

Data security

Is the robot in a unique data collection position? How is the user's data protected?

Screen can be glimpsed in more places than a fixed desk; device may be set down and left unattended or lost.		Auto-lock on inactivity or when set down; re-authentication to unlock; show only the current patient's data.	
PROBLEM	USER	ROBOT	SOLUTION
DEVICE	ROBOT	DEVICE	
A lost/stolen portable device is a bigger data exposure than a bolted-down terminal; Wi-Fi replaces a fixed connection.		Full-disk encryption; remote lock/wipe on report of loss; device-level network auth; wipe triggers only on explicit loss report, not routine connectivity drops.	



DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?

Think back to what you defined in the solution space canvas.

Precise tasks- Classification logic must be applied consistently every time, regardless of which nurse or shift; no personalization or variability that would undermine the "consistent, repeatable" advantage over unaided human judgement.

Connection to systems- Must pull directly from Mercer's EHR for patient context - this is the tablet's core advantage over a paper-based or purely manual process.

Interaction must stay minimal-touch/quick-glance so the precision advantage isn't offset by added time cost - a one-tap confirm/override preserves speed rather than trading accuracy for delay.

Ethical guidelines

What ethical considerations does the robot have?

Think back to what you defined in the ethics canvas.

No patient-facing interaction - patients aren't expected to touch or operate it, which keeps consent complexity lighter than Setting B's.

No voice interaction - supports data security by no verbal exposure of patient's info, & avoids the reliability issues that could produce a misleading confirmation under noise. One-tap confirm/override, with a mandatory logged reason for override - this directly implements accountability: a decision is always attributable to a named nurse.

Auto-lock on inactivity or when set down, requiring re-authentication - directly implements data security. Interface always shows the contributing features, never just a bare score - implements automation bias: the nurse engages with why, not just what.

Consent handled verbally as part of the existing triage conversation, not a new form - a practical answer to the data governance entry.